

Exploring User Privacy in Period and Fertility Tracking Applications

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Abstract

My doctoral research explores how to enhance the privacy practices and communications of period and fertility tracking apps to better meet user expectations and needs. Through a wide range of methods, my previous work explores how users perceive, understand, and respond to the privacy practices of these apps, particularly in light of recent legal shifts like the overturn of *Roe v. Wade*. Additionally, I examine how companies behind these apps articulate their privacy commitments, adapt to legal challenges, and address user concerns in their public communications. My future research will aim to bridge the gap between user expectations and app practices by conducting in-depth user interviews and a participatory design study to identify concrete pathways for enhancing privacy measures that resonate with user needs. My dissertation seeks to contribute to the HCI and CSCW fields by deepening current understanding of user privacy sensemaking and offering design and policy implications for enhancing privacy in period and fertility tracking apps.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Security and privacy** → **Human and societal aspects of security and privacy**.

Keywords

Period tracking, fertility tracking, privacy, personal informatics, reproductive health, collective privacy sensemaking, privacy communication, risk assessment, identifiability, anonymity, usability, overturn of *Roe v. Wade*

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1 Research Motivation and Background

In the digital age, the development of health technologies has transformed how individuals monitor and manage their health [43].

Among these technologies, period and fertility tracking applications like Clue and Flo have emerged as significant tools for millions worldwide seeking to understand their reproductive health better [1, 13, 44]. These apps collect sensitive health data related to menstruation and fertility from users, including menstrual cycle start and end dates, physical symptoms, and sexual activities, to make predictions about menstruation cycles and fertile windows [46]. This collection and usage of sensitive health data underscores pressing concerns about how such information is protected and utilized within these apps [13, 44].

Yet legal protections for period and fertility tracking data remain fragmented and uneven, creating a regulatory gap that exposes users to privacy risks [25, 29, 35]. For example, in the US, currently there are no comprehensive federal privacy laws and regulations to protect this special kind of data, resulting in data privacy risks in these apps [25, 29, 35]. In the UK and the EU, ambiguity also persists regarding whether and under what conditions period and fertility tracking data is treated as GDPR “special category” data [29]. Although GDPR provides heightened protections for sensitive personal data, the classification and treatment of period and fertility tracking app data can be unclear in practice, raising concerns that stricter regulatory obligations may not be consistently triggered or enforced for these apps [29, 40]. At the same time, legal and political shifts can further amplify both perceived and actual risks surrounding reproductive health data [5, 8, 31, 39, 40]. For example, in the U.S., the post-*Roe* environment has heightened fears that period and fertility tracking app data could be accessed or used in investigations or prosecutions, and legal scholars have warned about the potential use of such data in abortion-related cases [5, 31]. Beyond the U.S., restrictive reproductive health policies, criminalization, and social stigma in many regions worldwide can similarly shape how individuals evaluate the safety of sharing intimate reproductive health information [2]. Together, these dynamics create a complex privacy landscape that necessitates an in-depth examination of privacy concerns and expectations around period and fertility tracking apps.

The landscape of privacy in period and fertility tracking apps, has been shaped by a complex interplay of user concerns, legal challenges, and corporate practices. Previous research in HCI has examined the privacy policies and data practices of period and fertility tracking apps [14, 16, 19, 29, 37]. Legal scholars have further emphasized the inadequacies in current privacy regulations, highlighting the consequent privacy risks that users face [25, 35]. Notably, prior literature also reveals users’ heightened privacy concerns in period and fertility tracking data, exacerbated by significant legal changes such as the overturn of *Roe v. Wade* [12, 24, 27]. Despite these



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concerns, period and fertility tracking apps still inadequately protect sensitive data post-Roe, such as health, location, and personal details (e.g., name, email, and birth), with the potential for sharing with third-party authorities [8, 12, 26].

Despite extensive research on privacy issues in period and fertility tracking apps, important gaps remain in understanding how users make sense of privacy and what expectations they hold in practice. In addition, the relationships between users and other stakeholders—such as the companies behind these apps—remain underexplored, particularly amid rapidly shifting reproductive health governance. While heightened privacy concerns underscore the inadequacy of current privacy controls, few studies have focused on identifying effective, user-centered design solutions to address these issues. Therefore, it is essential to examine users' privacy concerns and management strategies in period and fertility tracking apps and to develop practical design interventions that better align with users' needs within this complex privacy landscape.

2 Research Objectives

My doctoral research aims to answer the critical question: In the evolving legal and social landscape, how can we enhance the privacy practices and communications of period and fertility tracking apps to better meet user expectations and needs? Specifically, the research is structured around three primary objectives: Objective 1: Holistically examining how users perceive, understand, and respond to privacy practices of period and fertility tracking apps, especially considering recent legal changes of *Roe v. Wade* (Study 1). Objective 2: Investigating how companies articulate their commitment to privacy, adapt to legal challenges, and engage with user concerns through their public communications (Study 2). Objective 3: Bridging the gap between user expectations and company practices, thereby identifying pathways to enhance privacy measures in ways that align with users' privacy needs (Studies 3–5).

My research is an ongoing effort aimed at addressing all the research objectives. My three complete papers [38–40] mainly address Objective 1 & Objective 2 by examining users' perception and sensemaking regarding privacy of period and fertility tracking apps, especially in the post-Roe context (Study 1) and how these app companies communicate their stance on privacy in response to landmark events like the overturn of *Roe v. Wade* (Study 2). I also conducted a usability testing study [41] of anonymity-preserving features of period and fertility tracking apps (Study 3) to address Objective 1 and Objective 3. Next, I will conduct an interview study (Study 4) and a participatory design study (Study 5) to further address Objective 3. Figure 1 demonstrates an overview of the five studies and the status of them.

3 Work to Date

3.1 Study 1: Investigating Collective Sensemaking of Privacy in Period and Fertility Tracking (Objective 1)

With the overturn of *Roe v. Wade*, the present privacy practices of period and fertility tracking apps pose even greater risks for users [17, 34, 45]. Although a limited number of studies have explored users' privacy concerns—such as the potential for prosecution and

Study 1: Users' privacy sensemaking of period and fertility tracking	Study 2: Apps' privacy communication addressing the overturn of Roe	Study 3: Usability of anonymity-preserving features of period and fertility tracking apps	Study 4: How period and fertility tracking app users protect their privacy	Study 5: Participatory design of privacy in period and fertility tracking apps
Completed	Completed	Completed	Proposed	Proposed
Objective 1	Objective 2	Objective 1&3	Objective 3	Objective 3

Figure 1: Overview of the five studies and the status of them

surveillance, particularly in states where abortion is criminalized [12, 27]—there remains a significant gap in understanding how users make sense of and respond to these risks as a community. Previous research has primarily focused on individual or group-based perspectives and collaborative privacy management [4, 9, 11, 21, 23]. However, the privacy risks stemming from the Roe overturn affect a broader demographic, including all women and people with uteruses. This is not just an individual concern but a societal issue, introducing additional legal risks and complexities to privacy in period and fertility tracking apps. Therefore, it is crucial to move beyond individual concerns and explore how people collectively make sense of these privacy risks, considering their broader societal implications.

In this study [39], through a thematic analysis [6] on online discussion data collected from Reddit, I identified three interconnected strands of people's collective sensemaking of privacy in period and fertility tracking apps: People collectively assess data privacy of period and fertility tracking apps with contextual information, such as app companies' official documents, historical actions, and business models; People also speculate about the privacy risks that the overturn of Roe might induce or intensify, such as the risks of prosecution, surveillance and harassment; perceiving the privacy risks, people brainstorm strategies to resist possible data surveillance and collaboratively protect their privacy. I revealed that collective sensemaking offers distinct advantages in navigating the complex landscape of privacy concerns post Roe. It allows individuals to collectively assess the potential risks and brainstorm means to resist surveillance, although their sensemaking is still affected and limited by data literacy, technology understanding, and the apps' opaque data practices. However, collective sensemaking could also amplify speculation and distrust due to ambiguities. People question the legality of period and fertility tracking data but also express grave concerns about the legal implications of it. Such legal risk compounds people's existing privacy concerns about their period and fertility tracking data, putting them in a further vulnerable position and perpetuating the oppression over the individual's body. I conclude with privacy design implications for privacy design in period and fertility tracking apps and contribute insights that could inform policymaking and legal perspectives.

3.2 Study 2: Exploring Period and Fertility Tracking App Companies' Privacy Communication Regarding the Overturn of *Roe v. Wade* (Objective 2)

The heightened privacy concerns of period and fertility tracking apps after the overturn of *Roe* have prompted some users to discontinue their use of these apps and encourage others to cease their usage as well [17], and some to seek alternatives that are more privacy focused [32]. To reassure their users regarding data privacy, some apps have issued public statements in the media [32, 45]. These public statements constitute an important source for understanding how companies address user privacy concerns and the privacy landscape in the post-*Roe* era. Previous research in HCI has investigated how period and fertility tracking companies communicate with their users about privacy through privacy policies [3, 14, 29, 37]. However, little attention has been paid to how companies use public statements to address historic events such as the overturn of *Roe* and communicate their privacy stance to users. Compared to privacy policies—which primarily function as legal documents—public statements are more accessible to the general public and often directly respond to user concerns and privacy-related events [7, 22]. As such, they offer a valuable yet underexplored data source for examining companies' privacy values and communication strategies. Building on this prior work, our study investigates how period and fertility tracking app companies communicate their privacy stances in response to historic privacy events, such as the overturn of *Roe*, through public statements.

In this study [38], I collected public statements released by period and fertility tracking apps from various sources, including company websites and official social media accounts. Our data analysis approach integrated thematic analysis [6] and discourse analysis [18] to gain a comprehensive understanding of how companies communicate their privacy stances through public statements. I identified nine recurring themes, such as community, empathy, brand identity, technological features, business models, and legal compliance. Our analysis shows how companies use discursive strategies and rhetorical devices—such as contrast, metaphor, hyperbole, and emotive language—to frame privacy as both a core value and an ethical responsibility, often embedding privacy commitments within operational and regulatory practices. These statements also address potential data requests, reflecting companies' attempts to balance user privacy with legal obligations. However, I found that such communications frequently lack transparency and clarity, which may undermine users' understanding of actual privacy protections. By situating companies' privacy stances within real-world contexts such as the post-*Roe* landscape, this study highlights public statements as a rich yet underexplored data source for examining shifting privacy norms, offering insights beyond those provided by privacy policies alone.

3.3 Study 3: Examining the Usability of Anonymity-Preserving Features in Period and Fertility Tracking Apps (Objective 1 & 3)

In the post-*Roe* landscape, concerns about reproductive health surveillance have intensified, making anonymity a critical privacy

need for users of period and fertility tracking apps [8, 26, 33]. Prior research shows that these apps engage in extensive data collection, opaque data-sharing practices, and, in some cases, insecure technical implementations, while users simultaneously express strong desires for privacy protection and anonymization [8, 29, 33]. Period and fertility tracking apps routinely collect highly sensitive reproductive health data alongside personally identifiable information (PII), such as names, email addresses, or payment details, heightening privacy risks within an expanding surveillance ecosystem [29, 33]. As a result, users fear that their data may be accessed, shared, or exploited by adversaries including government agencies, insurers, advertisers, and anti-abortion groups [8, 27, 30, 39]. These risks are reinforced by technical analyses revealing opaque cross-jurisdictional data flows and insecure data transmissions vulnerable to interception [14, 20, 27, 28]. In response, users have called for privacy features that support anonymity [8], and some period and fertility tracking apps (e.g., Flo and Cycles) have claimed to provide an “anonymous experience” [10, 15]. However, little is known about whether these anonymity-preserving features are usable in practice or how their usability shapes users' privacy perception and their ability to manage privacy risks.

In this study [41], I conducted a usability testing study of anonymity-preserving features in period and fertility tracking apps, examining how users interact with these features and how their usability influences users' privacy perceptions. I conducted 32 usability testing sessions across six systematically selected apps, asking participants to complete tasks focused on tracking period and fertility data without providing PII and, where possible, unlinking identity information post-registration. Using a mixed-methods analysis combining quantitative usability measures and qualitative thematic analysis, I examined both usability outcomes and users' experiences navigating anonymity-related designs. The findings show that the usability of anonymity-preserving features significantly influences users' perceived privacy risk, sense of control, trust, and overall usability of privacy settings. However, many apps rely on opaque, implicit, or fragmented designs that obscure anonymity options, impose burdensome workflows for avoiding data disclosure, and provide little feedback for users to evaluate the effectiveness of these features. As a result, even when anonymity-related features exist, poor usability undermines their protective potential and can amplify distrust. This study highlights anonymity as not only a technical property but a usability challenge, and contributes design implications for more transparent and privacy-by-design approaches in reproductive health technologies. Additionally, thematic analysis of the pre- and post-task interviews reveals how participants conceptualize personally identifiable information and the risks associated with it in perceived high-stakes contexts, such as using period and fertility tracking apps in the post-*Roe* legal landscape [40]. These interviews also shed light on users' broader privacy expectations regarding anonymity, encryption, and related protective features in these apps.

4 Expected Next Steps

With a foundational understanding of both user concerns and period and fertility tracking app companies' communication strategies, the next phase of my research will focus on privacy designs within

these apps to further address Objective 3 of identifying pathways to enhance privacy measures in ways that align with users' privacy needs.

4.1 Proposed Study 4: An Interview Study of How Period and Fertility Tracking App Users Protect Their Privacy (Objective 3)

Building on our previous investigation of users' sensemaking of privacy risks, app companies' privacy communications, and their current privacy features, this study aims to explore users' real-world experiences and strategies for protecting their privacy while using period and fertility tracking apps. Through in-depth qualitative interviews, I will examine how users understand, manage, and mitigate privacy risks in everyday settings, especially in light of heightened concerns following recent legal changes. The study will involve a diverse group of participants to capture a broad spectrum of privacy behaviors, motivations, and concerns. I will also ground participants' perceptions in recent technical analyses of period tracking apps (e.g., [20]). To ensure relevance to the apps used by our participants, I will also conduct a lightweight audit of commonly mentioned apps, examining their privacy features, permissions, and data-sharing behaviors.

This research will focus on users' practical strategies for safeguarding their personal data, their perceptions of the tracking methods they use, and how they navigate the privacy features currently available. Additionally, I will explore the factors influencing users' trust in these apps and their developers, including how privacy policies and features are communicated. By relating user perceptions to the observed privacy behaviors of the apps, the study will identify mismatches between user expectations and actual privacy practices. The findings will highlight the specific privacy challenges users face and identify which privacy features they find most valuable or lacking in current apps. This study is expected to uncover gaps between the privacy controls provided by app developers and the protections users expect, serving as a foundation for future design improvements in these digital tools.

4.2 Proposed Study 5: A Participatory Design Study to Improve Privacy Designs in Period and Fertility Tracking Apps (Objective 3)

After delving into the existing research landscape surrounding period and fertility tracking apps [3, 8, 14, 16, 19, 20, 29, 37], I identified a significant gap in the design and implementation of privacy features within these digital tools. Despite the widespread use and the sensitive nature of the data collected by these apps, current literature reveals a lack of user involvement in the design process, specifically concerning privacy protections [30]. This research is motivated by the urgent need to enhance user trust and safety in period and fertility tracking apps, as well as understanding what privacy features users expect. I plan to use the participatory design (PD) [42] method to gain a systematic and holistic understanding of how privacy designs in these apps can better support users and address their privacy needs. By adopting a PD approach, this study will directly involve users in the co-design of privacy features, enabling a systematic exploration of users' privacy needs,

expectations, and preferences in the context of period and fertility tracking apps. Specifically, the study will examine what types of data users are willing to share, under what circumstances they feel comfortable sharing such data, and what forms of privacy controls they find most intuitive and reassuring. It will also explore how users would like apps to communicate the current state of their data privacy within the interface, particularly in high-stakes contexts where reproductive health data may carry heightened legal, social, or surveillance risks. The design activities will be informed by prior literature as well as findings from my earlier studies [36, 39–41].

The PD workshop procedure includes recruiting participants who have experience or are interested in using period and fertility tracking apps with certain inclusion criteria, material preparation, exploration of participants' experiences, co-design activities, and evaluation through focus group discussions. The expected outcomes of this study will include a set of concrete design artifacts, such as privacy feature ideas, interface concepts, and low-fidelity prototypes, along with design recommendations grounded in users' lived experiences, privacy expectations, and concerns in high-stakes contexts. These outcomes will generate actionable guidance for developers and companies seeking to better align app design with user expectations. For example, the study will identify which privacy features users find most meaningful, how such features should be communicated within the interface, and what design strategies may make privacy protections more trustworthy, understandable, and usable in practice. By translating user input into concrete design recommendations and prototype concepts, this study aims to help bridge the gap between user expectations and current practices of these apps.

5 Contributions

The contributions of my doctoral research are multi-fold. First, it provides an in-depth analysis of user privacy perceptions, sense-making and expectations in the context of period and fertility tracking apps, enriching the existing literature on users' privacy concerns of these apps in the HCI field. Second, by investigating app companies' privacy communication and designs in response to legal changes, this research offers a critical examination of how companies adapt their privacy narratives and approaches in times of legal uncertainty. Third, through usability testing of privacy features and participatory design activities with users, this research aims to identify usability challenges and generate user-centered design recommendations for privacy-preserving features in period and fertility tracking apps. Finally, by integrating analyses of user perspectives, app practices, and design interventions, this work contributes both to academic discussions on privacy and reproductive health technologies and to practical guidance for app developers, policymakers, and legal stakeholders.

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